Sheet 62 Tue. 27 Apr. 2027 Synthesis problems spanning several areas of the curriculum

1. Warm-up · Flash Solve 5 $x - 3 = 2x + 9$.
2. Core · Problem A price undergoes a 20 % rise then a fixed discount of 1,000 Ar and is then 11,000 Ar. What was the initial price?
3. Core + · Method $f(x) = 2x + 1$ and $g(x) = -x + 7$. Find the intersection point of the curves then solve $f(x) < g(x)$.
4. Back · Flash A recipe for 4 people requires 250 g of flour. How much for 6 people?
5. Running thread · Flash A figure with area 4 cm ^{2} is enlarged by a scale factor of 3. What is the area of the enlargement?

Sheet 63 Thu. 29 Apr. 2027 Proof by contradiction and by case analysis

1. Warm-up · Flash What is the parity of $n(n+1)$ for an integer n?
2. Core · Case analysis Show that $n(n+1)$ is even for every integer n.
3. Core + · Contradiction Show by contradiction that the sum of a rational r and an irrational number a is irrational.
4. Back · Formula $g(x) = 2x + 5$. Image of -1? Preimage of 11?
5. Running thread · Formula → graph Plot the points with x-coordinates -2; -1; 0; 1; 2 of the curve of $f(x) = x^2$; give their y-coordinates.

Sheet 64 Mon. 3 May 2027 Logical connectives, contrapositive and counterexample in context

1. Warm-up · Flash Write the negation of " $x > 2$", then of " $x > 2$ or $x < -1$" .
2. Core · Method Write the contrapositive of "if n^2 is even, then n is even".
3. Core + · Counterexample Disprove: "every prime number is odd".
4. Back · True/false True or false? Justify. The perimeter of a square is proportional to the length of its side.
5. Running thread · Reading code What does $h(3)$? Write $h(x)$ as an expression.

Probability: conditioning and weighted trees

Sheet 65 Tue. 4 May 2027 Probability model and equally likely outcomes

1. Warm-up · Flash A fair coin is tossed three times. How many outcomes?
2. Core · Method Loaded die: $P(6) = 0.25$ and the other five faces are equally likely. Calculate $P(1)$.
3. Core + · Critical thinking "It will either rain or not rain tomorrow: so there is a one-in-two chance." Is this equally likely model justified?
4. Back · Flash Which shapes make up the net of a square-based pyramid?
5. Running thread · Translate A square with side x has each of its sides increased by 3. Express the increase in area as a function of x.

Sheet 66 Mon. 10 May 2027 Law of large numbers (simplified version): simulation

1. Warm-up · Flash Out of 100 tosses, 47 "heads". What is the frequency of "heads"?
2. Core · Interpret Why, over 10 tosses of a fair coin, can the frequency of "heads" move away from 0.5? What happens with 10,000 tosses?
3. Core + · True/false True or false? Justify. Tossing a fair coin 1,000 times, we get exactly 500 "heads".
4. Back · Problem Price incl. VAT $P(x) = 1{,}2\,x$ (20% VAT). Calculate $P(250)$ and the preimage of 480.
5. Running thread · Method Solve in $\mathbb{R}$: $\dfrac{2x-1}{3} = \dfrac{x+4}{2}$.

Sheet 67 Tue. 11 May 2027 Python simulation of a repeated random experiment

1. Warm-up · Flash $P(A) = 0.35$. Calculate $P(\overline{A})$.

2. Core · Reading code What does this program calculate? Approximately what value will it display?

```
from random import randint
compte = 0
for i in range(1000):
    if randint(1, 6) == 6:
        compte = compte + 1
print(compte / 1000)
```

3. Core + · Method A fair 12-sided die numbered 1 to 12. Probability of obtaining a number prime?

4. Back · Construction steps M lies on the perpendicular bisector of $[AB]$. What can be said about MA and MB? How can it be constructed with compasses?

5. Running thread · Recognise What kind of shape is the cross-section of a cube by a plane parallel to one of its faces?

Sheet 68 Thu. 13 May 2027 Conditional probability P $_A(B)$

1. Warm-up · Flash Of 25 half-boarders, 15 are girls. A half-boarder is chosen at random: probability that it is a girl?

2. Core · Method In a class of 30 pupils, 12 play football, of whom 5 also play handball. A pupil is chosen at random: calculate $P_F(H)$, F : "plays football", H : "plays handball".

3. Core + · Method In an equally likely model, $\text{Card}(A) = 50$ and $\text{Card}(A \cap B) = 20$. Calculate $P_A(B)$.

4. Back · Flash Put in ascending order: $\frac{7}{4}$, $-\frac{3}{2}$, 0,5.

5. Running thread · True/false True or false? Justify. A quadrilateral with two opposite sides parallel is a parallelogram.

Sheet 69 Tue. 18 May 2027 Conditional probability and two-way table

1. Warm-up · Flash In the following table, what is the total of the "sport" row?

	Girls	Boys	Total
Sport	18	22	…
No sport	12	8	20
Total	30	30	60

2. Core · Method A pupil is chosen at random from this table. Calculate $P(S)$, $P_F(S)$ and $P_S(F)$ (F : girl, S : does sport).

3. Core + · True/false True or false? Justify. In this table, $P_F(S) = P_S(F)$.

4. Back · Flash Calculate $5^0 + 2^{-1}$.

5. Running thread · Method $ABCD$ is a parallelogram with $A(1;1)$, $B(5;2)$, $C(6;5)$. Determine D.

Sheet 70 Thu. 20 May 2027 Weighted trees: construction

1. Warm-up · Flash What is the sum of the probabilities on the branches leaving the same node?

2. Core · Method In a population, 30% have a disease M. A test is positive (T) for 90% of those who are ill and for 5% of those who are not. Complete the branch probabilities of the tree.

3. Core + · Translate Translate into symbols: "among those who are ill, 90 % test positive".

4. Back · Flash Calculate $\frac{2}{3} \div \frac{4}{9}$.

5. Running thread · Problem Three lighthouses flash every 6, 8 and 9 seconds. They light up together at $t = 0$. When is the next simultaneous flash?